CLINICAL PROGRAMMING BOOTCAMP • MODULE 13 • CAPSTONE

Map it yourself

A study you have never seen — DEF-01 — from raw to SDTM, on your own

Everything you have built so far had a worked notebook beside it. This one does not. That is the point: mapping data you have never seen is the actual job.

Deliverable: 13_capstone_DEF01_SKELETON.sas → six SDTM domains → verify_capstone.sas

YOUR ASSIGNMENT

DEF-01 — a Phase 2 study in Type 2 diabetes

Drug X 50 mg once daily vs placebo · 6 subjects · 2 sites · 12 weeks

Site 01 is in Canada (metric); site 02 is in the USA (US customary units). Each subject is screened, dosed at Baseline, and followed to a Week 12 visit. The raw data is five EDC-style CSVs — one per form — with the same messy realities you have been cleaning up all course.

Build one domain from every observation class

DM	**Special Purpose**	demographics + reference dates
EX	**Interventions**	study-drug exposure
AE + SUPPAE	**Events**	adverse events + AETRTEM
VS · LB	**Findings**	vital signs + labs

Disposition and Concomitant Medications are out of scope — five domains is a full end-to-end pass without being a slog.

FOUR THINGS THAT WILL BREAK A BLIND COPY-PASTE

Your ABC-01 code is the start — but not the finish

1 · UNIT CONVERSION

VS

Site 02 collects weight in lb and temperature in °F. Keep the collected value in --ORRES; CONVERT to kg / °C in --STRESN.

2 · DOSING INTERRUPTION

EX

Subject 02/003 has two dosing periods. Keep both records; the gap is the interruption. RFSTDTC = first dose, RFENDTC = last.

3 · PARTIAL DATE

AE

One screening AE is dated to a month only (2024-02). Keep the partial; AESTDY is null; never impute a day. AETRTEM is still N.

4 · ABNORMAL LABS

LB

Many labs are HIGH — that is the diabetes, plus one ALT signal. Derive LBNRIND. Do NOT invent adverse events for them.

None of these is a curveball — each is a realistic situation you have the tools for. They are here because ABC-01 never happened to contain them.

HOW TO WORK

Attempt first, then verify — the whole course in one loop

1. **Read the dictionary** def01_data_dictionary.md describes every raw field and every trap. Start here, not in the code.
2. **Fill in the skeleton** 13_capstone_DEF01_SKELETON.sas gives you the raw reads and a TODO per domain. Write the mapping.
3. **Check your row counts** DM 6 · EX 7 · AE 7 · SUPPAE 7 · VS 96 · LB 32. A wrong count localises the problem fast.
4. **Run verify_capstone.sas** It compares every value to the reference and reports MATCH or FAIL per domain. Aim for six MATCHes.
5. **Fix and re-verify** A FAIL shows the first differing lines. Read them, fix the mapping, run again — exactly how real QC works.

There is no worked notebook — but there IS a reference. "Attempt first, then compare" is how you have learned every domain; now it is the whole task.

LOOK HOW FAR YOU HAVE COME

Ten days, from zero to a full SDTM package

Foundations	what a trial is, why CDISC exists, the CRF→SDTM→ADaM→submission path
The raw truth	raw data is never ISO; mixed formats, codes, free text, wide vs tall
Every domain	DM, DS, AE, SUPPAE, CM, EX, VS, LB — one from every observation class
The cross-cutting skills	USUBJID, --DY (no Day 0), --SEQ, Controlled Terminology, the transpose
Quality & submission	conformance checks, SUPPQUAL & RELREC, define.xml, XPT v5, traceability
And now — transfer	a study you have never seen, mapped end to end, on your own

THIS IS NOT THE END OF THE PIPELINE

Where SDTM sits, and what comes after

CRF / EDC — the site collects data → **SDTM** — tabulation — what you now build → **ADaM** — analysis-ready datasets, derived from SDTM → **TLFs** — the tables, listings & figures in the study report → **Submission** — define.xml + datasets + guides → the regulator

You are here — and it is the foundation everything downstream stands on.

ADaM is built FROM SDTM; the analysis tables are built from ADaM; the submission wraps it all in define.xml. Every layer above depends on the tabulation data being right — which is why so much of this course was about getting it right and proving it.

The natural next step: ADaM

ADaM (Analysis Data Model) takes your SDTM and adds the derivations analysis needs — treatment-emergent flags applied, change-from-baseline computed, one analysis-ready row per record. If SDTM is "what happened", ADaM is "what the statistician analyses". It is the obvious place to go next.

WHERE TO GO FROM HERE

The map, now that you can read it

The real standards	CDISC.org for the SDTMIG; NCI-EVS for Controlled Terminology. What you learned maps directly onto the published documents — they will now read as familiar.
The real tools	Pinnacle 21 Community (free) to validate; real MedDRA and WHODrug for coding. You wrote the checks by hand — now you know what the tools do.
The next standard	ADaM for analysis datasets, then TLFs. SDTM is the foundation; these are the floors above it.
Keep the instincts	attempt then verify · fail loudly · never invent data · trace every value back. Those outlast any one standard.

You started ten days ago not knowing what a domain was. You can now map a clinical study to SDTM and prove it. Go build DEF-01 — and welcome to clinical programming.